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SPECIAL REPORT · ENERGY & TECHNOLOGY

The Long Ignition Nuclear Fusion and the Road to the Grid

An assessment of where fusion energy actually stands in 2026 — the physics that has been proven, the machines being built, the money behind them, and the honest distance still left to commercial power.



EDITOR

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The Long Ignition

Nuclear fusion and the road to the grid

TL;DR

Fusion crossed a genuine physics threshold in December 2022, when the National Ignition Facility got more energy out of a laser-driven pellet than the lasers put in. Nothing since has repealed that result, and it has unlocked an unprecedented wave of private capital and political will.

But "ignition" in a lab and "electricity on the grid" are different problems. ITER, the flagship public tokamak, will not even attempt full fusion until 2039. The best-funded private companies — Commonwealth Fusion, Helion, TAE, Tokamak Energy — are now racing engineering timelines rather than physics unknowns, with Commonwealth already filing to connect a 400 MW plant to the US grid and Helion contracted to sell Microsoft power from 2028.

Cumulative private investment has passed roughly \$10–13 billion, regulators in the US and UK have started writing rules specifically for fusion rather than fission, and yet, as of 2026, no company is within a year of putting fusion electrons on a real grid. The most honest summary is that fusion has stopped being a physics question and become an engineering and capital one — which is a real form of progress, just not the one the headlines imply.

01 Introduction and Scope

For most of the last seventy years, nuclear fusion has served as a standing joke about the gap between scientific promise and engineering reality: always thirty years away, regardless of the decade in which the joke was told. That joke has aged badly since 2022. This essay sets out, plainly and without the marketing gloss that increasingly surrounds the sector, where fusion energy actually stands in the second half of 2026 — what has been proven, what is under construction, who is paying for it, and what a realistic timeline to commercial electricity looks like.

The scope is deliberately narrow. This is not a physics primer on the strong nuclear force, nor a stock-picking guide to fusion equities. It is a survey of the state of the art across the three tracks that matter: the large public programmes (ITER and its national equivalents), the inertial-confinement laser programme centred on the U.S. National Ignition Facility, and the fast-moving private sector now racing to turn decades of plasma physics into a power plant. Each is at a

different point on the curve from "the physics works" to "the plant is profitable," and conflating them is the single most common source of misleading fusion journalism.

02 What "Ignition" Actually Means

Fusion releases energy by forcing light atomic nuclei — usually isotopes of hydrogen — to fuse into a heavier nucleus, the reverse of the fission reaction used in today's nuclear plants. Because two positively charged nuclei repel each other ferociously, doing this on purpose requires temperatures several times hotter than the core of the Sun, contained for long enough, and at high enough density, for fusion reactions to outrun the energy lost from the system. The three requirements — temperature, density, confinement time — are traded off against one another in what is known as the Lawson criterion, and the entire sixty-year history of fusion research is really a history of pushing different combinations of those three variables closer to the line where the reaction pays for itself.

Two engineering philosophies have dominated the chase. Magnetic confinement uses powerful superconducting magnets to hold a ring of plasma away from any physical wall, sustaining the reaction for as long as the magnets and heating systems allow — this is the tokamak and stellarator family, and the approach behind ITER, EAST, SPARC, and most private tokamak companies. Inertial confinement instead compresses a tiny frozen fuel pellet so violently, usually with an array of lasers, that fusion occurs in a few billionths of a second, before the fuel has time to fly apart — the approach used at NIF and pursued commercially by firms such as Pacific Fusion and Xcimer. Both are, in effect, trying to solve the same equation from opposite ends: one buys time by containing a large low-density plasma indefinitely; the other buys density by compressing a tiny target so fast that time barely matters.

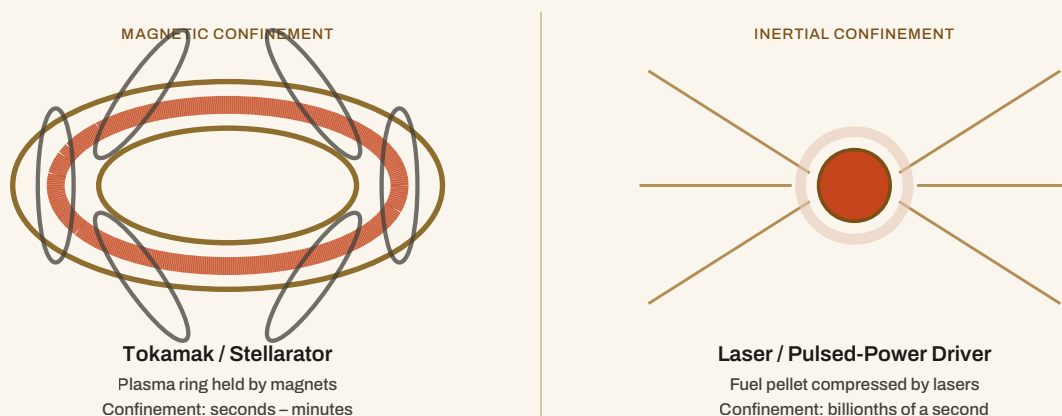


Figure 1. The two dominant confinement philosophies. Magnetic confinement sustains a low-density plasma for seconds to minutes inside a magnetic bottle; inertial confinement compresses a high-density pellet for billionths of a second. Commercial designs increasingly borrow instrumentation and materials science from both.

The milestone that changed the public conversation came from the inertial side. In December 2022, the National Ignition Facility reached a target that had eluded researchers for six decades: a single laser shot yielded more fusion energy than the laser energy used to drive it, with the fuel pellet releasing roughly 3.15 megajoules of fusion energy from about 2.05 megajoules of laser input. It is worth being precise about what that result did and did not prove. It confirmed, for the first time under laboratory conditions, that a fusion reaction can be self-sustaining enough to exceed the energy delivered to the fuel. It did not prove that a laser fusion power plant is economically viable: NIF's own lasers draw on the order of 300 megajoules of electricity from the wall to produce those 2.05 megajoules of laser light, a "wall-plug" efficiency problem that commercial inertial-fusion companies are now trying to solve with entirely different laser and pulsed-power architectures.

03 The Public Backbone: ITER and the National Programmes

ITER — the International Thermonuclear Experimental Reactor under construction in Cadarache, southern France — remains the largest and most expensive fusion experiment ever attempted, a collaboration spanning the European Union, the United States, China, Russia, Japan, South Korea, and India. It is a machine built to prove sustained, self-heating "burning plasma" at a scale no private company can currently finance alone, and it will feed its magnet, heating, and remote-maintenance engineering directly into every serious commercial reactor design that follows it. It is also years behind its original schedule: a 2024 replanning exercise pushed first plasma out to the mid-2030s, with operation beginning around 2035 and full deuterium-tritium fusion not attempted until 2039. Critically, ITER is a physics and engineering test bed rather than a power station, and it is not designed to put any electricity onto the grid even once running at full strength.

China has pursued a parallel and, on one specific metric, faster-moving programme. Its Experimental Advanced Superconducting Tokamak (EAST) in Hefei — dubbed the "artificial sun" in domestic coverage — has repeatedly broken its own record for sustaining a stable, high-confinement plasma, reaching a sustained 1,066 seconds — roughly eighteen minutes — of steady, high-confinement plasma operation in January 2025, at temperatures exceeding 100 million degrees Celsius. Duration is not the same as net energy gain — the announcement, as independent reporting has noted, made no claim about whether the shot produced more energy than it consumed — but pulse length and plasma stability are exactly the engineering problems a commercial tokamak has to solve, and EAST's results feed directly into China's planned CFETR demonstration reactor, the country's own bridge between ITER-class physics and a working power plant.

ON THE FUNDING SIDE

Public fusion budgets remain modest next to the scale of the ambition. U.S. federal fusion energy science funding sits at roughly \$806 million for FY2026, of which around a fifth is directed to the

ITER contribution; the UK, by contrast, has committed a record £2.5 billion over five years to its own STEP programme and supporting infrastructure, including a new fusion-dedicated supercomputer at Culham. These sums still look small against the private capital now flowing into the sector — a genuine reversal from fusion's historically government-dominated funding model.

04 The Private Race: From Physics Experiment to Engineering Sprint

The defining story of fusion since 2022 is not a new physics result — it is the arrival of serious private capital chasing engineering timelines the public programmes were never designed to meet. Depending on whose count is used, the private fusion sector has attracted somewhere between \$10 billion and roughly €13 billion in cumulative investment, an eightfold increase since 2020, spread across more than seventy companies worldwide, the large majority of it concentrated in the United States.

€13bn+

CUMULATIVE PRIVATE
FUSION INVESTMENT
WORLDWIDE, 2026

70+

PRIVATE FUSION
COMPANIES NOW
IDENTIFIED GLOBALLY

~53%

SHARE OF GLOBAL PRIVATE
FUNDING HELD BY U.S.
FIRMS

400MW

COMMONWEALTH FUSION'S
PROPOSED VIRGINIA GRID
CONNECTION

Four companies illustrate the range of the field. **Commonwealth Fusion Systems**, spun out of MIT and backed by more than \$2 billion from investors including Bill Gates and Google, is building on high-temperature superconducting magnets that generate far stronger fields than ITER's in a much smaller machine, allowing its SPARC demonstration device to target similar plasma performance at a fraction of ITER's size. In 2026 the company took the step that most sharply signals commercial intent rather than pure research: it filed an application to connect its planned ARC power plant — the roughly 400-megawatt Fall Line Fusion Power Station in Virginia — directly to PJM, the largest electricity grid in the United States, targeting delivery in the early 2030s. It is the first time any developer has asked to connect a commercial fusion plant to a major grid.

Helion Energy has taken the most aggressive public timeline in the industry and backed it with a real commercial contract. In 2023 it signed what was billed as the fusion industry's first power purchase agreement of any kind: a commitment to supply Microsoft with at least 50 megawatts of electricity from 2028, with Constellation Energy managing the transmission. Helion uses a pulsed, magneto-inertial approach — closer in spirit to a linear collider than a tokamak — and by mid-2026 had broken ground on the plant site in Chelan County, Washington, while continuing to test successive prototypes. The company remains a minority-case bet: most independent physicists regard a 2028 in-service date as extremely tight, and Helion's own backers, including

OpenAI's Sam Altman as its largest private investor, have been candid that the agreement carries financial penalties if the date slips.

Alongside these two headline projects sits a broader field: **TAE Technologies**, pursuing an aneutronic hydrogen-boron approach that would avoid neutron activation and tritium-breeding problems altogether if it can be made to work; **Tokamak Energy** in the UK, developing compact spherical tokamaks; and inertial-confinement entrants such as **Pacific Fusion**, whose Sirius pulsed-power prototype passed a 3,000-shot milestone in 2026 in partnership with Lawrence Livermore, alongside Zap Energy's Z-pinch machines and General Fusion's magnetized target approach. Major energy companies — Chevron, Eni, Shell among them — have taken strategic stakes across this field, a sign that incumbents now see fusion as a live technology risk worth hedging rather than a curiosity to ignore.

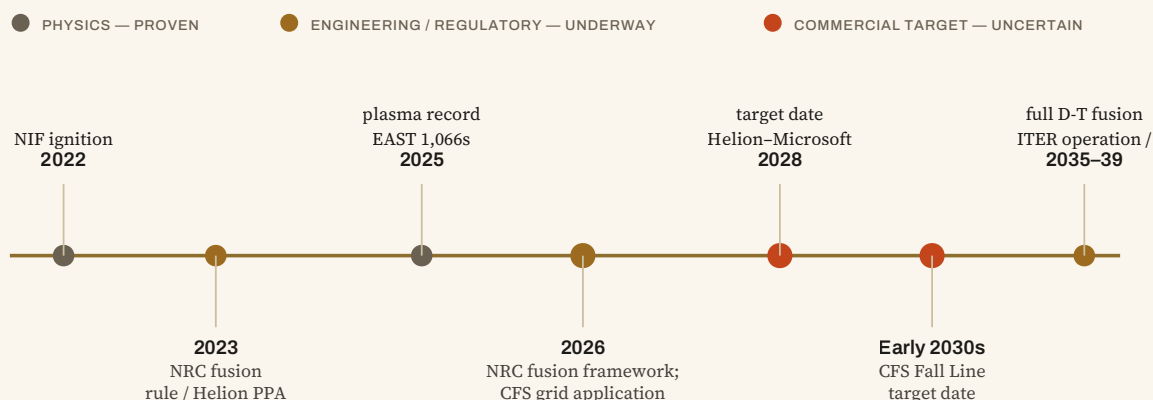


Figure 2. Selected fusion milestones, 2022–2039. Physics milestones (grey) have already been met; engineering and regulatory milestones (brass) are underway; commercial targets (plasma-orange) remain the least certain dates on the chart.

05 The Regulatory Shift

Regulation has quietly become one of the more consequential fusion stories of the last three years, precisely because it signals that governments now expect commercial machines to exist. In the United States, the Nuclear Regulatory Commission voted in 2023 to regulate fusion devices under the same lighter-touch "byproduct material" framework used for particle accelerators and medical isotopes, rather than the framework built for fission reactors — a formal acknowledgment that a tokamak or laser chamber which cannot melt down poses a fundamentally different risk profile to a fission core. That principle was carried further in February 2026, when the NRC published its proposed Regulatory Framework for Fusion Machines in the Federal Register, opening formal public comment on rules written specifically for fusion rather than adapted from

fission. The UK has moved in the same direction through its own fusion strategy funding, treating regulatory readiness as a deliverable alongside the hardware itself.

This matters commercially in a way that is easy to underrate: a lighter, purpose-built regulatory pathway removes years of licensing uncertainty that would otherwise sit on top of the engineering risk private investors are already taking. It is one of the clearest examples of policy moving ahead of proven commercial technology in the sector's history — a bet, in effect, that the underlying physics and safety case are now solid enough to legislate for.

Magnetic and inertial approaches compared

DIMENSION	MAGNETIC CONFINEMENT (TOKAMAK / STELLARATOR)	INERTIAL CONFINEMENT (LASER / PULSED-POWER)
Leading example	ITER, EAST, SPARC (Commonwealth Fusion)	National Ignition Facility, Pacific Fusion's Sirius
Confinement time	Seconds to minutes, continuously heated	Billionths of a second, single pulsed shot
Maturity	Most mature; decades of tokamak operating data	Ignition proven once (Dec 2022); repeatability and rate still developing
Core private challenge	Magnet manufacturing at scale, neutron-resistant materials	Laser/driver efficiency, pellet manufacturing rate, repetition rate
Commercial posture	Grid-connection applications already filed (Commonwealth Fusion)	Still largely at demonstrator / prototype stage commercially

06 What Still Stands Between the Lab and the Grid

None of the preceding progress removes a set of engineering problems that have nothing to do with plasma physics and everything to do with materials, supply chains, and economics. A commercial deuterium-tritium reactor has to breed its own tritium fuel inside the reactor wall, since the isotope barely exists in nature and world supplies are minute; it has to survive a continuous bombardment of high-energy neutrons that degrade structural materials over the plant's lifetime in ways still being characterised; and it has to manufacture high-temperature superconducting magnet tape, laser optics, or pulsed-power components at industrial volumes and industrial cost, not laboratory scale. None of these is a fundamental barrier in the way that achieving net energy gain once was. All of them are the unglamorous, capital-intensive work that separates a demonstration reactor from a power plant that a utility will actually buy.

It is this gap that explains the wide spread in credible timelines. One recent industry survey found that the large majority of fusion companies — 84 percent, by one count — expect commercial fusion by the 2030s, an optimistic, self-selected view from the firms with the most to gain from that belief. Independent assessments are more cautious, converging on a blunter picture: as of 2026, no commercial fusion plant has put electricity on a grid, no company is within a year of doing so, and the industry's most aggressive early milestones have slipped, even as the underlying physics, engineering, and capital trajectory have all moved meaningfully forward. Both statements are true at once, and holding them together is the only honest way to read the sector.



A note from the editor. I have tried, throughout this piece, to keep the "genuinely proven" column separate from the "confidently promised" column. Readers of the Monetary History series will recognise the instinct: promissory notes are not the same thing as gold in the vault, whether the note is denominated in dollars or in megawatts.

— Summary

Nuclear fusion crossed a real and durable scientific threshold in December 2022, when the National Ignition Facility demonstrated for the first time that a fusion reaction could release more energy than was delivered to the fuel. That result, together with a decade of steady progress in superconducting magnet technology, has reframed fusion from an open physics question into a capital-intensive engineering programme — and capital has responded, with more than \$10 billion in cumulative private investment now backing over seventy companies worldwide.

The public programme, led by ITER, remains the indispensable proving ground for the reactor-scale engineering no private company can finance alone, even as its own timeline has slipped to the late 2030s. China's EAST tokamak has pushed plasma-duration records further and faster than almost any other single facility. And the private sector has begun making commitments that look, for the first time, like genuine commercial intent rather than research ambition: Commonwealth Fusion's grid-interconnection application for a 400 megawatt plant, and Helion's contracted 2028 power sale to Microsoft, chief among them.

Set against that progress is an unglamorous but decisive residue of engineering problems — tritium breeding, neutron-resistant materials, and manufacturing fusion components at industrial rather than laboratory scale — that no funding round or press release removes. The most defensible view in 2026 is that fusion has earned the right to be taken seriously as a 2030s energy technology, without yet having earned the right to a firm delivery date. Readers should treat any specific year attached to "commercial

fusion" — 2028, 2031, 2035 — as a target under active pursuit, not a certainty already banked.

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